

Design and Optimization of a Configurable Pixelated Antenna Using a Genetic Algorithm

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Abstract—Modern antenna design standards are evolving, and there has been a significant interest in algorithm-driven antenna design. Predefined, conventional structures are often implemented and optimized through optimization techniques. However, recent trends show that optimized pixelated antennas can outperform classical designs due to greater design flexibility. This study proposes a novel approach towards designing a pixelated antenna with no conventional shape, which could be configured to resonate at any frequency using a genetic algorithm. To achieve this, a radiating structure composed of an 8x8 grid of metallic pixels was generated, with each pixel being independently controlled by a binary parameter. This structure was then mirrored across both the X and Y axes for symmetry. As this was a binary optimization task, a genetic algorithm was used to find the optimal pixel configuration that achieves resonance at a desired frequency while minimizing return loss. The antennas were simulated in CST Microwave Studio while the Genetic Algorithm was run on MATLAB to generate the antenna parameters. The generated structures show excellent performance with return loss (S11) values of -18.05dB at 2.45GHz, -28.93dB at 3GHz, -16.49dB at 4GHz, -28.78dB at 5GHz, and -22.34dB at 6GHz, respectively. A further fine-tuning process was also explored. The process was shown to be able to efficiently generate new antenna structures as per user demand.

Keywords—Pixelated Antenna, Genetic Algorithm, Binary Optimization Problem, CST Microwave Studio, MATLAB

I. INTRODUCTION

Microstrip antennas are an essential part of the modern telecommunication systems. Unlike traditional antennas, they are small and lightweight, making them an excellent choice for use in microwave frequencies. However, designing a microstrip antenna structure can be tedious, requiring much trial and error. Thus, it often takes a long time to design antennas that meet a specific performance criterion, as finding the optimum value of all the antenna parameters is time-consuming, and the optimization process can be a daunting task. To mitigate this issue, an algorithm-driven approach towards parametric optimization can be implemented, which could accelerate the design process, resulting in time savings and also minimizing errors [1].

Many have implemented different algorithms to optimize the parameters of an antenna so that proper resonance/performance is obtained. The most common way to do this is to design a microstrip antenna with sub-par performance and then optimize it through various means. W. A. Awan et al. in their work, designed a wideband antenna with notching characteristics where they used a parametric sweep to generate results and used a genetic algorithm (GA) to optimize the parameters [2]. A similar work is proposed by M. Boudjerda et al. where they designed a simple square patch

antenna, where the parameters were optimized using a genetic algorithm [3]. Researchers are also using machine learning or deep learning algorithms to improve time-saving antenna design performance [4] [5]. However, these traditional antenna design approaches still rely on a starting antenna structure that needs to be manually designed, resulting in a structure that may be limited by predefined geometry and design conventions.

A more flexible, algorithm-driven approach could be that of a pixelated antenna, which is gaining recent popularity. In such antennas, a pixel is a unit geometry of the radiating patch, which is tiled grid-like and can be populated with metal or left unpopulated, exposing the substrate [6]. These antennas are characterized by their highly customizable geometric configurations, where the radiating surface is subdivided into a grid of discrete, controllable pixel elements, resulting in the feasibility of implementing unconventional designs and offering far greater design flexibility compared to regular structures. This approach offers a more flexible design space, allowing the exploration of unconventional structures and geometries that would be challenging to conceive through traditional design methods [7]. However, the vast number of possible configurations presents a significant challenge for designers, as it becomes computationally infeasible to evaluate every possible design manually. To solve this problem, automated optimization approaches can be implemented, such as particle swarm optimization [8] [9], and particularly evolutionary algorithms [10] could be used as this counts as a binary optimization problem.

Pixelated antennas have emerged as a novel approach to achieving high-performance designs. M. A. Ullah et al. in their work, designed a low-profile dual-band antenna with a pixelated defective ground structure for IoT applications [11]. D. Mair et al., in their works, implemented a pixelated antenna using shifted cross-shaped elements and automatically optimized it with the help of evolutionary algorithms. for 5G applications [12]. D. Mair et al. also conducted another study where they investigated the performance of pixelated antennas based on their geometrical parameters [13]. All of these researches highlight the implications of using pixelated antenna structures. Very little work has been done where pixelated antennas were optimized by an algorithm for reconfigurability.

This paper proposes a novel algorithm-driven approach towards designing pixelated antennas. A pixelated antenna was first created in CST Microwave Studio software. A connection between CST and MATLAB was then established using a toolbox that is publicly available as a *GitHub* repository [14]. MATLAB was then used to implement a genetic algorithm (GA) along with goals and constraints. Now MATLAB could change the antenna parameters and as the

antenna structure was designed in such a way that changing parameters changes the antenna pixel structure, any structure could be created based on the requirements. The antenna shape was then altered using a genetic algorithm to make it resonate at any given target frequency and the results were assessed by MATLAB. As pixelated antennas having n number of pixel elements have 2^n number of possible configurations, the implemented genetic algorithm was found to be effective in traversing this search and determining the optimum parameter combination for any input goal.

Section II of the paper explains the proposed methodology of this work. All the fine details related to the structural design and algorithm can be found here. Section III contains the result analysis for this study. Section IV is the conclusion where a brief summary of the entire work is presented.

II. METHODOLOGY

This section of the paper describes the methodology used to implement a genetic algorithm based optimization to a pixelated antenna. Fig. 1 shows the block diagram of the proposed work.

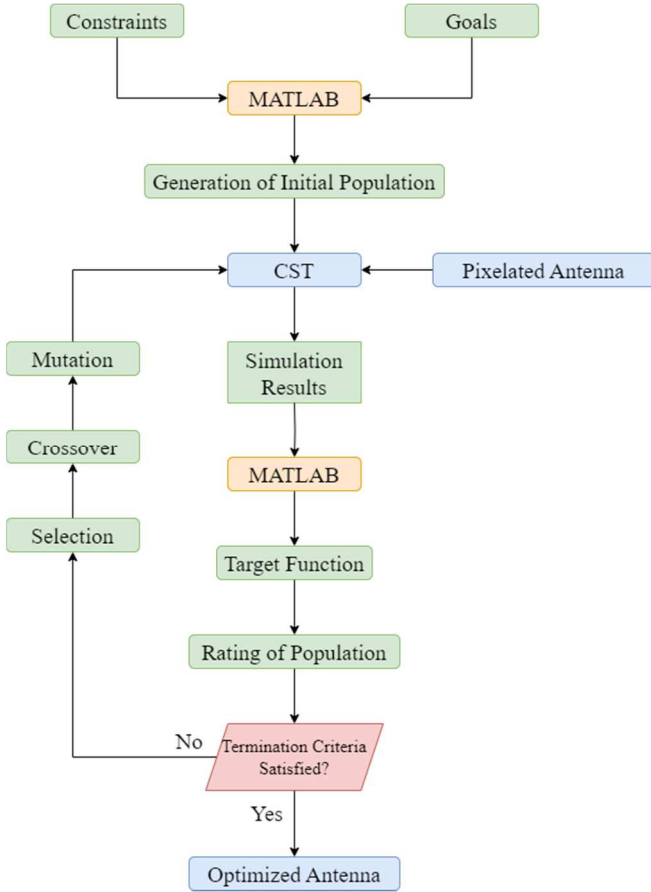


Fig. 1. Block diagram of the proposed work.

A. Antenna Design as a Binary Optimization Problem:

The process begins with designing the antenna structure, which will then be optimized using a sophisticated and precise tool- the genetic algorithm. It ensures that the antenna design is optimized to its full potential. CST Microwave Studio software was used to implement the antenna. To begin the designing process, a substrate was created using FR-4 as

the substrate material with an epsilon of 4.3. A grounding plane was added beneath the substrate, which has a thickness of $0.01mm$ and made perfect electric conductor (PEC). A feedline was created on top of the substrate from the edge to feed the antenna elements with power, which has the same material and thickness as the grounding plane. It has a width of $3mm$ and length of $9mm$. After this, the radiating patch was designed. All these measurements can be found in TABLE I.

TABLE I. ANTENNA PARAMETERS LIST

Parameter Name	Parameter Values
FW (Feed Width)	$3mm$
FL (Feed Length)	$9mm$
HC (Height of Copper)	$0.01mm$
HS (Height of Substrate)	$0.8mm$
SL (Length of Substrate)	$50mm$
SW (Width of Substrate)	$50mm$
P1-P64 (Pixel Parameters)	0 or 1 (Boolean)

To design the radiating patch, the antenna structure was divided into four quadrants, centering at the origin (0,0) point. The patch was initially designed on the top-right quadrant as an 8×8 pixel array, with each pixel measuring $2 \times 2mm^2$. The equations used to design the pixels are given below:

$$X \text{ dim.: } m * 2 \text{ to } (m - 1) * 2 \text{ for } 1 \leq m \leq 8 \quad (1)$$

$$Y \text{ dim.: } n * 2 \text{ to } (n - 1) * 2 \text{ for } 1 \leq n \leq 8 \quad (2)$$

$$Z \text{ dim.: } 0 \text{ to } HC, \text{ translated to } (-HC - HS) * p \quad (3)$$

Here, m is the row index, and n is the column index of the pixel matrix created on the top right quadrant. The height of each element is HC but is translated in such a way that depending on the value of p , the element will either stay on top and act as a radiating patch or get merged with the ground plane. As there are 8×8 or a total of 64 initial elements, p is denoted by $P1$ to $P64$ for all 64 elements, respectively.

To ensure symmetry and uniformity in the design, this initial configuration was mirrored across both axes, resulting in a patch structure that spans all four quadrants of the substrate. Fig. 2 shows this entire process.

Now that the pixelated structure is defined, the individual metal patches can be selectively turned on (kept as a radiating element) or turned off (merged with the grounding plane). This can be adjusted by altering the parameters $P1$ to $P64$, each responsible for each pixel element on the substrate. These 64 parameters can now be modified to give the antenna any shape desired, depending on the performance requirements. The primary objective now is to optimize the arrangement of these pixel elements to achieve resonance at the specified target frequency. To achieve this, a genetic algorithm-based optimizer is implemented to adjust the pixel configuration to minimize the return loss (S_{11}) at the desired frequency, ensuring efficient impedance matching and maximizing the antenna's radiation performance. Fitness of each configuration is evaluated based on the return loss such

that minimizing it improves the antenna's performance at resonance.

B. GA Encoding Scheme:

To form the chromosomes to be used with the genetic algorithm, the variables $P1$ to $P64$ were used, as they represent the antenna's pixelated structure. As each of these variables corresponds to a specific pixel on the substrate, they can only take a value of either 0 or 1. So, the variables were constrained to be integer only with an upper bound of 1 and a lower bound of 0, ensuring the turn-on/turn-off nature of the pixels. The initial population to be used with the algorithm is randomly generated, with each individual in the population representing a unique binary chromosome pattern and, thus, a unique antenna structure. The population was initialized with a maximum population count of 50 individuals, which ensures sufficient variety in the binary search space. Changing this chromosome pattern now changes the antenna structure.

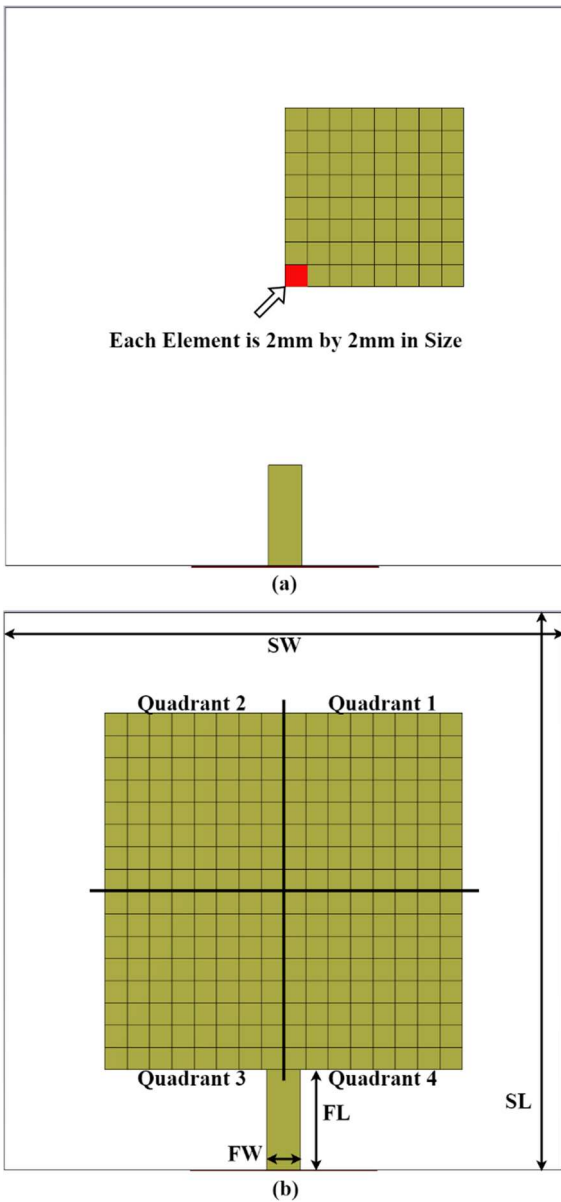


Fig. 2. Pixelated Antenna Structures, (a) initial design, (b) final design with all quadrants and sizes marked.

C. Fitness Evaluation:

These individuals are then sent to CST Microwave Studio, where they update the antenna structure. After this, the CST Solver was run to generate the results. As the $S11$ Parameter was the primary interest for this work, the $S11$ parameter is now collected from CST via MATLAB. From the results, the target frequency or query frequency was checked, and the value of return loss was assessed. As this generally results in a negative value, the fitness function or objective function can be defined as:

$$res = 20 * \log_{10}(|S|) \quad (4)$$

Where S is the value of return loss at the query frequency and res being the return loss in db , which needs to be minimized.

After all the individuals were evaluated, their results were used to generate the next generation of individual pool. The elite count was set to 2, meaning that out of all the individuals of the current generation, two were guaranteed to survive into the next generation. Fitness scaling was set to "rank-based," which scales the raw fitness scores to determine the probability of selection. A stochastic uniform selection function was used to select individuals for mating, ensuring that better individuals are more likely to be selected. During mating, a scattered crossover function was used, which randomly selects bits from two parent individuals to create offspring. The crossover fraction was set to be 80%, meaning 80% of the next generation of the population came from the crossover operation. After this step, a mutation operation was added to ensure that diversity was maintained within the population and help the algorithm explore a more expansive search space. The uniform mutation function was used because the chromosomes have binary encoding, which flipped bits in the chromosome to introduce diversity. The mutation rate was set to be adaptive, which changes as the algorithm runs longer. It starts with a higher value initially, encouraging exploration and avoiding premature convergence. As the epochs pass, the mutation rate is decreased to fine-tune the solutions more effectively.

D. Termination Criteria:

The optimization process keeps running until one of the termination criteria is met. The first default criteria used was *stall on function tolerance* where the algorithm stops when there is no significant improvement in the best fitness value over several generations. The default value is 10^{-6} , implying that the algorithm would stop in a certain number of generations if the improvement between generations were below the tolerance value. The second one was *stall on generations*, where the algorithm would stop when no improvement was seen in the best solution after a specific number of generations. The default value is 50. Another extra stopping criterion was implemented, which was *max number of generations*, and was set to 10. This would terminate the optimization cycle after ten generations. This was found to be adequate for the problem at hand.

By leveraging this entire process, an optimized antenna structure was generated for any given goal or target frequency. In this work, five antennas were generated that had good return loss ($S11$) at their target frequencies. These will be explored in the next section.

III. RESULT ANALYSIS

This section explores all the generated designs and their corresponding S_{11} parameters. It also shows a fine-tuning process that can be added after antenna design generation to further fine-tune the antenna's performance.

A. Antenna Optimization for 2.45 GHz Resonance

The process was first used to generate an antenna structure that could resonate at 2.45GHz. The generated structure can be seen in Fig. 3. As the initial pixel structure was mirrored two times, the generated structure is also symmetric about both axes. The simulated S_{11} parameter can be seen in Fig. 4. Fig. 5 shows the best individual selected by the genetic algorithm, resulting in the optimized structure.

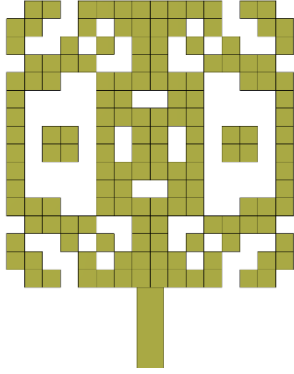


Fig. 3. Pixelated antenna structure optimized for 2.45GHz.

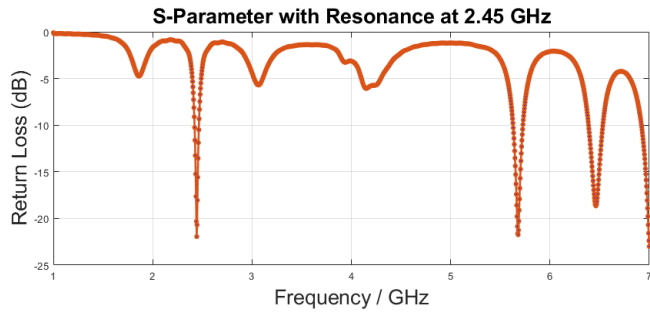


Fig. 4. S_{11} Parameter of antenna, optimized for 2.45GHz.

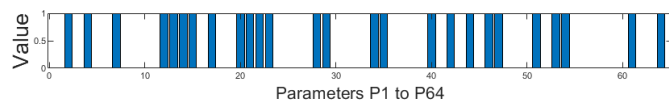


Fig. 5. Optimized parameter values (P1-P64) for 2.45GHz.

Fig. 4 also shows that the return loss at 2.45GHz is $-18.05db$. The antenna is slightly miss-tuned, which further fine-tuning can easily solve. Built in optimizers in CST Microwave Studio can be used to find optimal value for all the other parameters that are tunable. This finds the optimum values for the remaining parameters unaltered by the genetic algorithm. This tuning process is described below for the structure generated for 5GHz.

B. Antenna Optimization for 3 GHz Resonance

The antenna structure generated to resonate at 3GHz can be seen in Fig. 6. Its corresponding simulated S_{11} Parameter can be seen in Fig. 7. The best individual selected by the genetic algorithm to achieve resonance at 3GHz can be seen in Fig. 8.

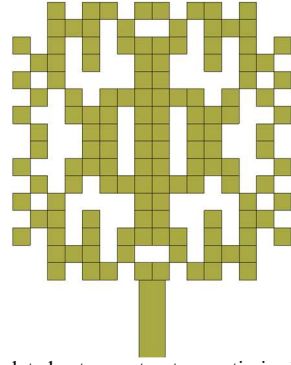


Fig. 6. Pixelated antenna structure optimized for 3GHz.



Fig. 7. S_{11} Parameter of antenna, optimized for 3GHz.

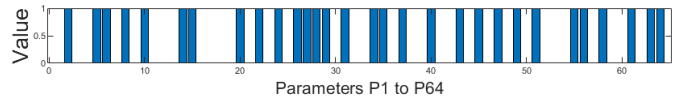


Fig. 8. Optimized parameter values (P1-P64) for 3GHz.

Fig. 7 shows that the return loss at 3GHz was found to be $-28.93dB$. The resonance frequency was found to be quite close to 3GHz.

C. Antenna Optimization for 4 GHz Resonance

Fig. 9 shows the antenna structure generated for 4GHz. Fig. 10 shows the resultant S_{11} Parameter, and Fig. 11 shows the best individual which generated the structure.

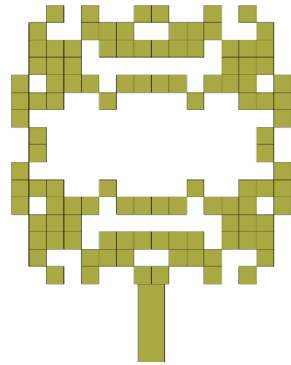


Fig. 9. Pixelated antenna structure optimized for 4GHz.



Fig. 10. S_{11} Parameter of antenna, optimized for 4GHz.

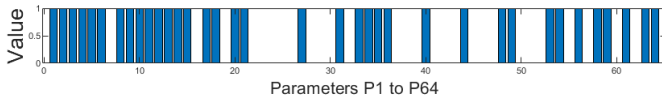


Fig. 11. Optimized parameter values (P1-P64) for 4GHz.

Fig. 10 shows that the return loss at 4GHz was found to be $-16.49dB$.

D. Antenna Optimization for 5 GHz Resonance

Fig. 12 shows the antenna structure generated for resonance at 5GHz. Its simulated S11 Parameter can be seen in Fig. 13. The best individual that generated the antenna structure can be seen in Fig. 14.

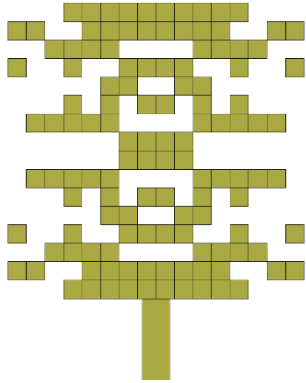


Fig. 12. Pixelated antenna structure optimized for 5GHz.

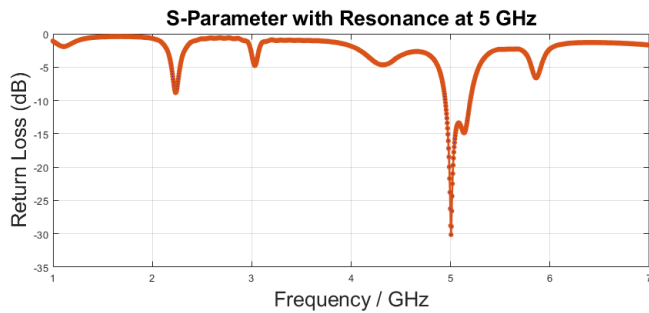


Fig. 13. S11 Parameter of antenna, optimized for 5GHz.

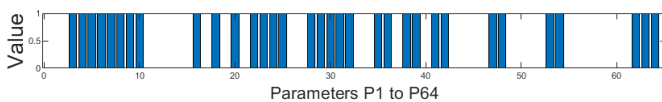


Fig. 14. Optimized parameter values (P1-P64) for 5GHz.

Fig. 13 shows that the return loss at 5GHz was found to be $-28.78dB$.

E. Antenna Optimization for 6 GHz Resonance

Fig. 15 shows the antenna structure generated for resonance at 6GHz. The resulting S11 Parameter is shown in Fig. 16. The best individual generating the structure is shown in Fig. 17.

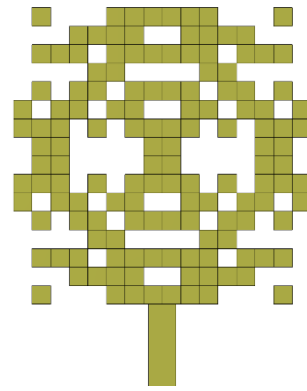


Fig. 15. Pixelated antenna structure optimized for 6GHz.

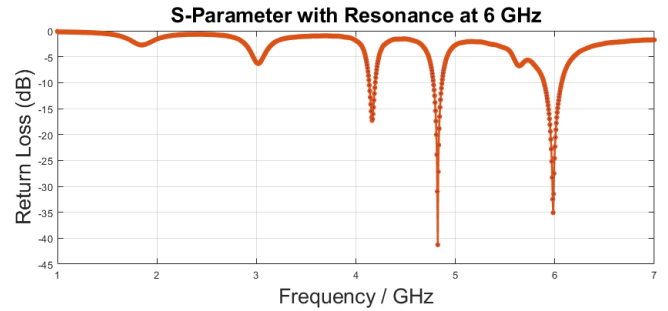


Fig. 16. S11 Parameter of antenna, optimized for 6GHz.

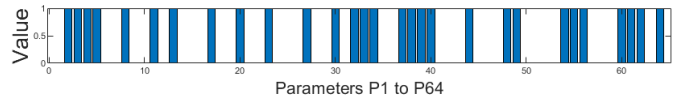


Fig. 17. Optimized parameter values (P1-P64) for 6GHz.

Fig. 16 shows that the return loss at 6GHz was found to be $-22.34dB$.

F. Further Fine-Tuning:

All the generated structures can be further refined via built-in tools in CST Microwave Studio, such as parametric sweep and optimizers. The Trust Region Framework optimizer was used to tune the antenna's performance further. It was given all the variables that could be fine-tuned and given a goal of minimizing return at 5GHz. The updated parameters can be seen in TABLE II.

TABLE II. ANTENNA PARAMETERS LIST

Parameter Name	Old Values	New Values
FW (Feed Width)	3mm	3.52mm
FL (Feed Length)	9mm	8.8893mm
SW (Width of Substrate)	50mm	50.1583mm

To prove that these values are optimal, parametric sweeps can be done to check the change in the antenna performance with changing parameter values. This can be seen in Fig. 18.

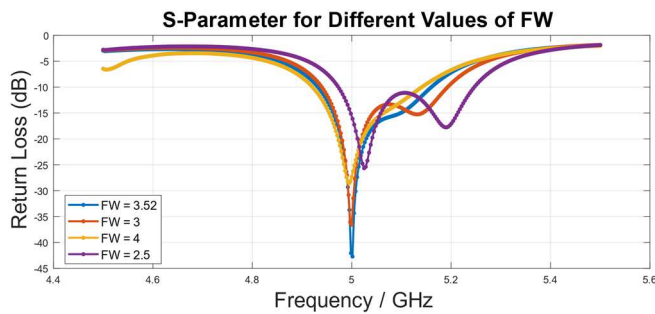


Fig. 18. Parametric analysis of changing return loss with respect to changing feed width for resonance at 5GHz.

Fig. 18 shows that the optimum value for feed width is 3.52mm. Increasing or decreasing the value results in worse performance at resonance frequency. It can also be seen that the updated return loss at 5GHz is -42.39dB . So, using this approach, antenna structures can be designed to resonate at any frequency and can also be further fine-tuned to increase performance.

IV. CONCLUSION

This study presents a novel approach towards designing pixelated antennas using a genetic algorithm. The initial grid-like structure of antenna pixel elements was modified using a GA optimizer to generate unconventional antenna structures that has a return loss of -18.05dB at 2.45GHz, -28.93dB at 3GHz, -16.49dB at 4GHz, -28.78dB at 5GHz, and -22.34dB at 6GHz. The generated structure for 5GHz was further fine-tuned to achieve a return loss of -42.39dB . This fine-tuning process could be applied to other results as well. This highlights the potential of pixelated antennas to outperform conventional designs in terms of flexibility. Also, the ability to optimize the antenna structure automatically, without needing trial and error, increases efficiency and time savings. Future work could focus on incorporating other metrics like gain and efficiency and also dynamic reconfigurability.

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